

WELCOME

Physics 1 [Fall 2023 - 2024]

**Department of Physics
Faculty of Science & Technology (FST)
American International University-Bangladesh**

COURSE: PHYSICS 1 (PHY 1101)

SEMESTER: Fall [2023-2024]

CREDIT: 3 CREDIT HOURS

MARKS DISTRIBUTION

ATTENDANCE AND PERFORMANCE: 10 (10%)

ASSESSMENTS (QUIZZES) : BEST ONE OUT OF TWO : 20 (20 %)

PRESENTATION : 20 (20%)

MIDTERM ASSESSMENTS (COUNT ALL): 50 (50%)

TOTAL = 100 POINTS/MARKS

Outline up to Mid term

Reference Book: Fundamentals of Physics (10th Edition)

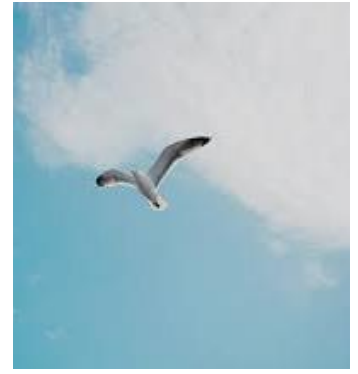
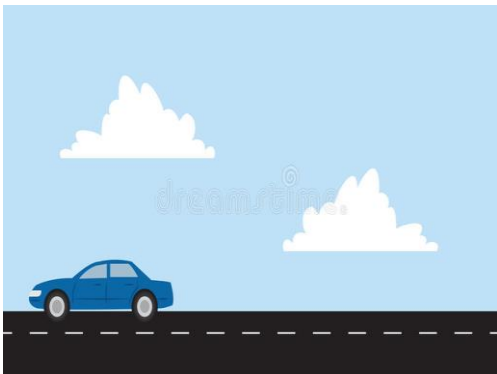
Written by Halliday, Resnick and Walker

Book chapter no	Chapter name
4	Motion in Two and Three Dimensions
5	Force and Motion-I
6	Force and Motion-II
7 and 8	Kinetic Energy and Work And Conservation of Energy
9	Center of Mass and Linear Momentum
10	Rotation
11	Rolling, Torque, and Angular Momentum

LESSON 1

BOOK CHAPTER 4

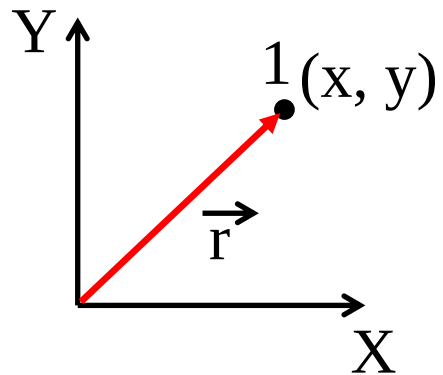
Motion in Two and Three Dimensions



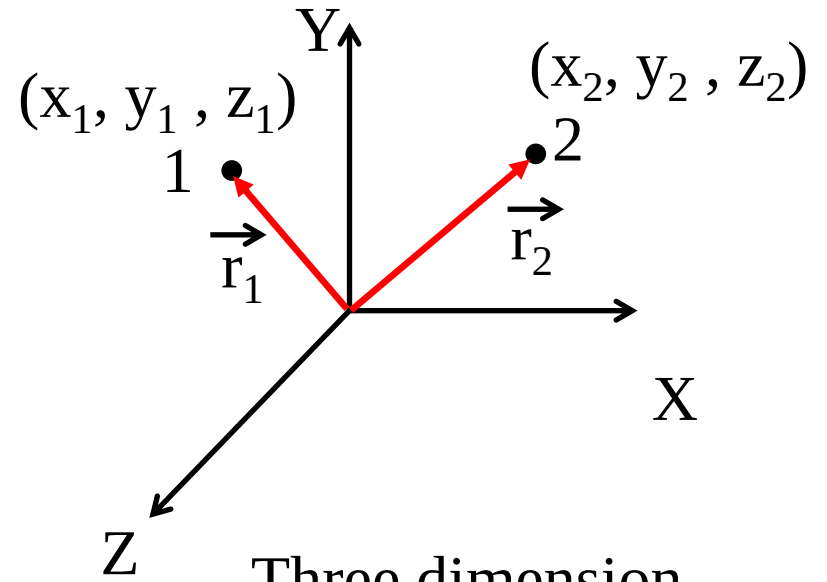
Outline of Lesson 1

- Position and Displacement
 - Average Velocity and Instantaneous Velocity
 - Average Acceleration and Instantaneous Acceleration
-

Position:



Two dimension

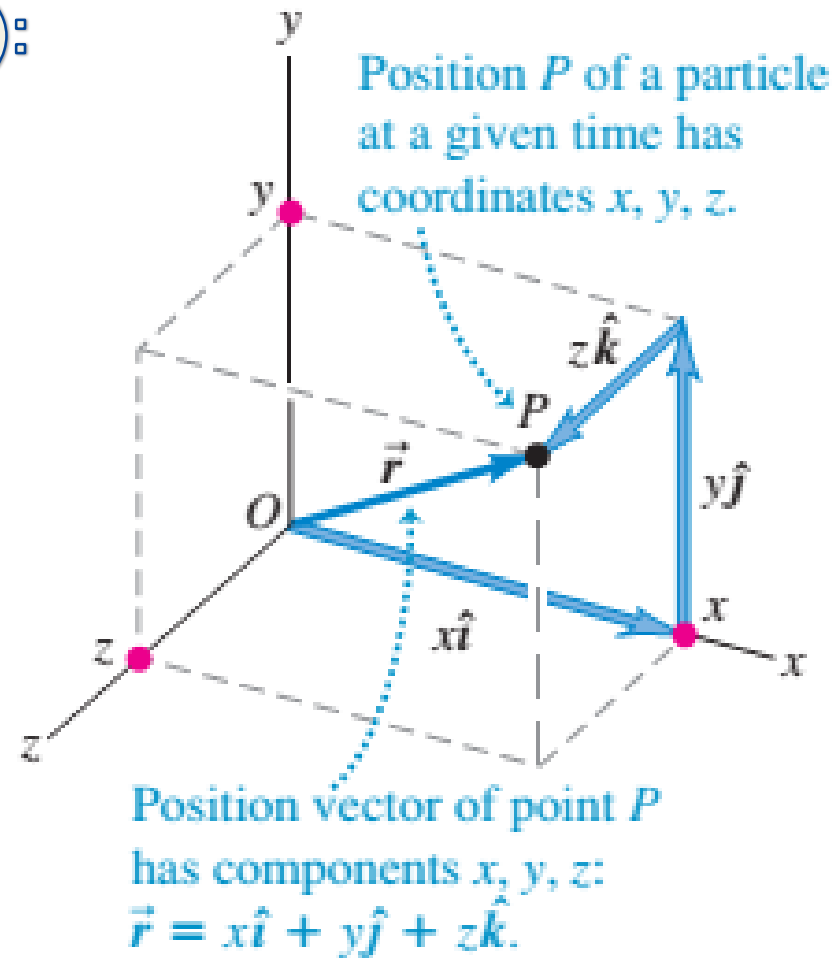


Three dimension

Position Vector (three-dimension):

To describe the *motion* of a particle in space, we must first be able to describe the particle's *position*. Consider a particle that is at a point P at a certain instant. The **position vector** $\vec{r}$ of the particle at this instant is a vector that goes from the origin of the coordinate system to the point P (as shown in the figure). The Cartesian coordinates x , y , and z of point P are the x -, y -, and z -components of vector $\vec{r}$. Using the unit vectors we can write

$$\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$$



Position Vector and Displacement Vector:

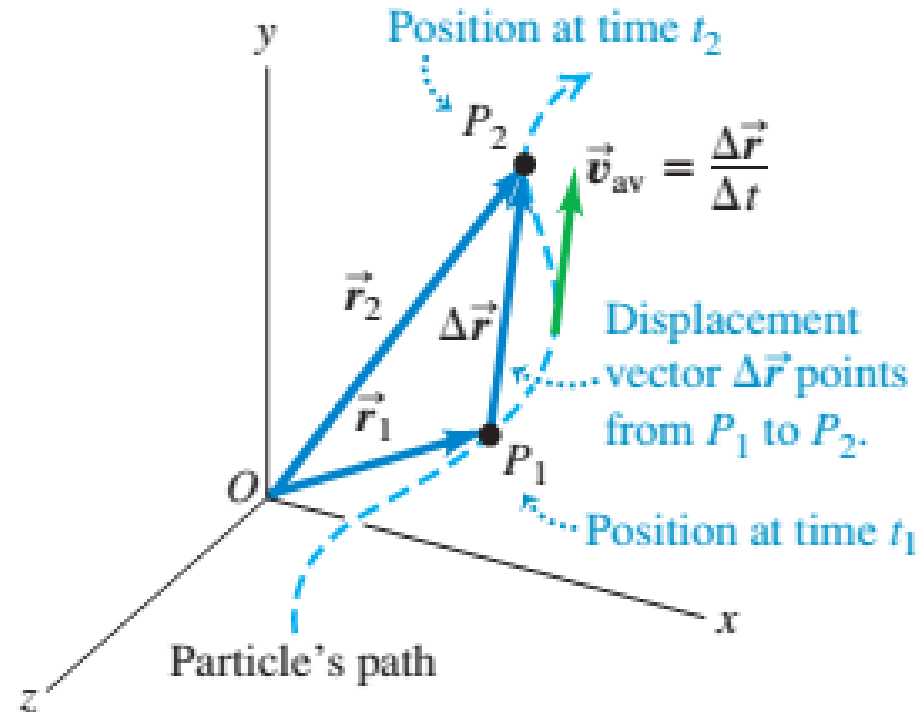
During a time interval Δt the particle moves from P_1 , where its position vector is $\vec{r}_1$ to P_2 , where its position vector is $\vec{r}_2$. The change in position (the displacement) during this interval is

$$\Delta\vec{r} = \vec{r}_2 - \vec{r}_1$$

$$\Delta\vec{r} = x_2\hat{i} + y_2\hat{j} + z_2\hat{k} - (x_1\hat{i} + y_1\hat{j} + z_1\hat{k})$$

$$\Delta\vec{r} = (x_2 - x_1)\hat{i} + (y_2 - y_1)\hat{j} + (z_2 - z_1)\hat{k}$$

$$\Delta\vec{r} = \Delta x\hat{i} + \Delta y\hat{j} + \Delta z\hat{k}$$



Average Velocity and Instantaneous Velocity:

If a particle moves through a displacement $\Delta\vec{r}$ in a time interval Δt , then its **average velocity** $\vec{v}_{avg}$ is

$$\vec{v}_{avg} = \frac{\vec{r}_2 - \vec{r}_1}{t_2 - t_1} = \frac{\Delta\vec{r}}{\Delta t}$$

Instantaneous velocity (simply, velocity $\vec{v}$) is the limit of the average velocity as the time interval approaches zero, and it equals the instantaneous rate of change of position with time. That is

$$\vec{v} = \lim_{\Delta t \rightarrow 0} \frac{\Delta \vec{r}}{\Delta t} = \frac{d\vec{r}}{dt}$$

The *magnitude* of the vector $\vec{v}$ at any instant is the *speed* of the particle at that instant. The *direction* of $\vec{v}$ at any instant is the same as the direction in which the particle is moving at that instant.

Note: At every point along the path, the instantaneous velocity vector is tangent to the path at that point.

□ Create a particle's position vector as a function of time and evaluate its (instantaneous) velocity vector.

$$\vec{r}(t) = x\hat{i} + y\hat{j} + z\hat{k}$$

We have the definition of velocity vector,

$$\vec{v} = \frac{d\vec{r}}{dt}$$

$$\vec{v} = \frac{d}{dt} (x\hat{i} + y\hat{j} + z\hat{k}) = \frac{dx}{dt}\hat{i} + \frac{dy}{dt}\hat{j} + \frac{dz}{dt}\hat{k} = v_x\hat{i} + v_y\hat{j} + v_z\hat{k}$$

The **magnitude** of the instantaneous velocity vector $\vec{v}$ —that is, the speed—is given in terms of the component v_x , v_y and v_z by the Pythagorean relation:

$$|\vec{v}| = v = \sqrt{v_x^2 + v_y^2 + v_z^2}$$

The adjacent Figure shows the situation when the particle moves in the xy -plane. In this case, z and v_z are zero. Then the speed (the magnitude of $\vec{v}$) is

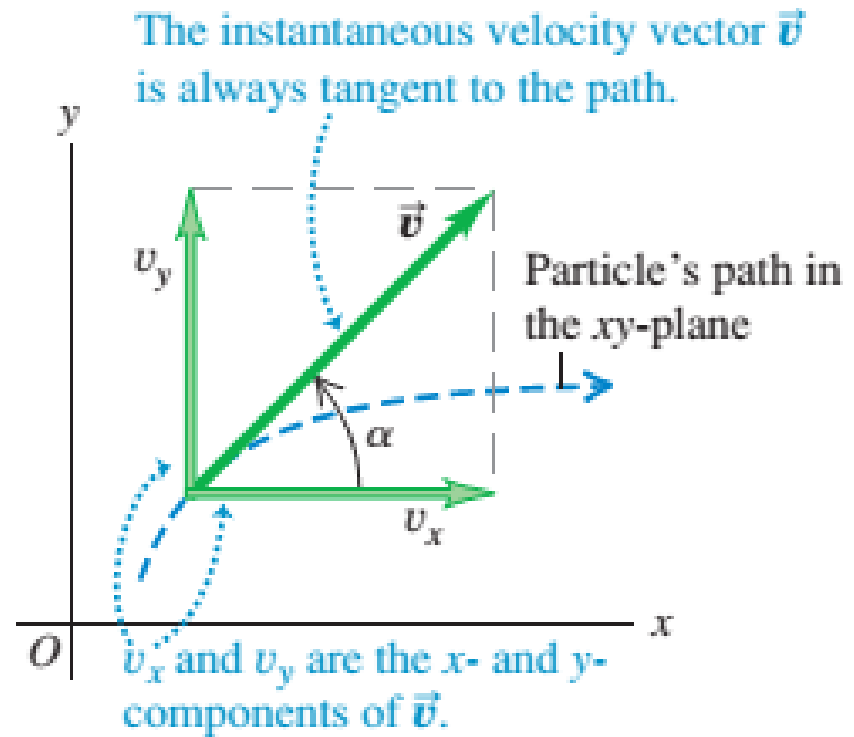
$$v = \sqrt{v_x^2 + v_y^2}$$

The **direction** of the instantaneous velocity is given by the angle α (the Greek letter alpha) in the figure.

$$\tan \alpha = \frac{v_y}{v_x}$$

And

$$\alpha = \tan^{-1} \frac{v_y}{v_x}$$



If a body's (or particle's) velocity changes from $\vec{v}_1$ to $\vec{v}_2$ in time interval Δt , its average acceleration during Δt is

$$\vec{a}_{avg} = \frac{\vec{v}_2 - \vec{v}_1}{t_2 - t_1} = \frac{\Delta \vec{v}}{\Delta t}$$

If Δt approaches to zero about some instant, then in the limit $\vec{a}_{avg}$ approaches the **instantaneous acceleration** (or **acceleration**) at that instant; that is,

$$\vec{a} = \lim_{\Delta t \rightarrow 0} \frac{\Delta \vec{v}}{\Delta t} = \frac{d\vec{v}}{dt}$$

□ Create a particle's velocity vector as a function of time and evaluate its (Instantaneous) acceleration vector.

$$\vec{v}(t) = v_x \hat{i} + v_y \hat{j} + v_z \hat{k}$$

$$\vec{a} = \frac{d\vec{v}}{dt} = \frac{d}{dt} (v_x \hat{i} + v_y \hat{j} + v_z \hat{k}) = \frac{dv_x}{dt} \hat{i} + \frac{dv_y}{dt} \hat{j} + \frac{dv_z}{dt} \hat{k}$$

$$\vec{a} = a_x \hat{i} + a_y \hat{j} + a_z \hat{k}$$

Problem 3 (Book chapter 4)

A positron undergoes a displacement $\Delta\vec{r} = 2\hat{i} - 3\hat{j} + 6\hat{k}$, ending with the position vector $\vec{r} = 3\hat{j} - 4\hat{k}$, in meters. What was the positron's initial position vector?

Answer:

We have $\Delta\vec{r} = \vec{r} - \vec{r}_1$

$$\vec{r}_1 = \vec{r} - \Delta\vec{r} = 3\hat{j} - 4\hat{k} - (2\hat{i} - 3\hat{j} + 6\hat{k}) = 3\hat{j} - 4\hat{k} - 2\hat{i} + 3\hat{j} - 6\hat{k}$$

$$\vec{r}_1 = -2\hat{i} + 6\hat{j} - 10\hat{k}$$

Problem 13 (Book chapter 4)

A particle moves so that its position (in meters) as a function of time (in seconds) is $\vec{r} = \hat{i} + 4t^2\hat{j} + t\hat{k}$. Write expressions for (a) its velocity and (b) its acceleration as functions of time.

Answer:

We have
$$\vec{v} = \frac{d\vec{r}}{dt}$$

$$\vec{v} = \frac{d}{dt}(\hat{i} + 4t^2\hat{j} + t\hat{k}) = 0 + 8t\hat{j} + \hat{k} = 8t\hat{j} + \hat{k}$$

Again, we have
$$\vec{a} = \frac{d\vec{v}}{dt}$$

$$\vec{a} = \frac{d}{dt}(8t\hat{j} + \hat{k}) = 8\hat{j} + 0 = 8\text{ m/s}^2\hat{j}$$

Do by yourself

1. [Chap 4 - problem 2]: A watermelon seed has the following coordinates: $x = 5.0$ m, $y = 8.0$ m, and $z = 0$ m. Find its position vector (a) in unit-vector notation and as (b) a magnitude and (c) an angle relative to the positive direction of the x axis. (d) Sketch the vector on a right-handed coordinate system. If the seed is moved to the xyz coordinates (3.00 m, 0 m, 0 m), what is its displacement (e) in unit-vector notation and as (f) a magnitude and (g) an angle relative to the positive x direction?
2. [Chap 4 - problem 7]: An ion's position vector is initially $\vec{r} = 5\hat{i} - 6\hat{j} + 2\hat{k}$, and 10 s later it is $\vec{r} = -2\hat{i} + 8\hat{j} - 2\hat{k}$, all in meters. In unit vector notation, what is its $\vec{v}_{avg}$ during the 10 s?
3. [Chap 4 - problem 11]: The position of a particle moving in an xy plane is given by $\vec{r} = (5t^3 - 5t)\hat{i} + (6 - 7t^4)\hat{j}$, with $\vec{r}$ in meters and t in seconds. In unit-vector notation, calculate (a) $\vec{r}$, (b) $\vec{v}$, and (c) $\vec{a}$ for $t = 2.00$ s.
4. [Chap 4 - problem 14]: A proton initially has $\vec{v} = 4\hat{i} - 2\hat{j} + 3\hat{k}$ and then 4.0 s later has $\vec{v} = -2\hat{i} - 2\hat{j} + 5\hat{k}$ (in meters per second). For that 4.0 s, what are (a) the proton's average acceleration $\vec{a}_{avg}$ in unit vector notation, (b) the magnitude of $\vec{a}_{avg}$, and (c) the angle between $\vec{a}_{avg}$ and the positive direction of the x axis?

Thank you